

Q11 charger with a Shelly X module

Test documentation and interface mapping: connection, serial protocol, data points, results, open questions

1 Purpose and scope

AmperePoint is evaluating the replacement of the Tuya WBR3 Wi-Fi module in the Q11 charger (factory product Q21_11kW_with_OTA, control board X2Q322_K_V4) with a Shelly X module. This document records what was tested on a real Q11 between 24 and 28 September 2026: how the module is connected, what the controller sends on the serial link, how its data points are mapped, which tests passed, and which questions remain open. It is written for the factory's R&D engineers.

Result in one sentence

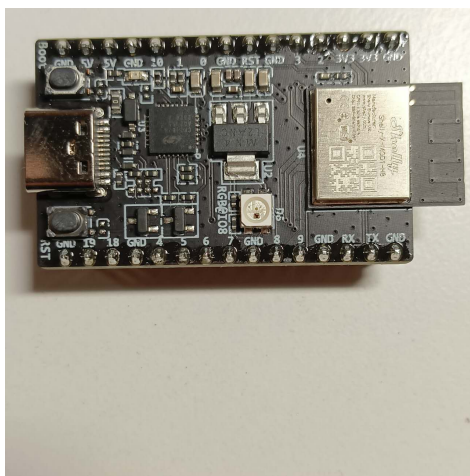
The Q11 controller works with the Shelly module **without any change to the controller firmware**. The Shelly module speaks the Tuya MCU serial protocol at 9600 baud; four connections (3.3 V, GND, RXD, TXD) are sufficient. Charging on/off, current limit, work state, connection state, temperature, phase data and fault bits were exchanged in both directions during about ten hours of frame-level logging on 25 September 2026 (10:51–20:48).

In scope	Details
Hardware	module site U10, pads actually used by the board, controller pins, power supply of the module from the board
Serial link	Tuya MCU protocol as observed on the wire: start-up handshake, routine traffic, write behaviour, example frames
Data points	all 17 data points of the product definition: type, encoding, whether the controller reports them, how they are mapped in the Shelly product
Tests	control and status in both directions, operation without cloud, time delivery, local integration layer (short)
Out of scope	
Cloud and app	the Shelly cloud, the Shelly app screens and the AmperePoint back-end are not described here; nothing in them requires anything from the controller

2 Equipment

Item	Details
Charger	AmperePoint Q11, 11 kW three-phase portable charger. Control board X2Q322_K_V4 (260703). Controller: LQFP64 microcontroller with sticker "X2Q3110W OTA V22". Tuya module WBR3 desoldered from site U10.

Item	Details
Prototype module	Shelly DevKit M1 v1.1 (Espressif reference board) carrying the Shelly-X-MOD1-H8 module (ESP32-C3 core, Shelly firmware 2.0.0-beta11), see Fig. 1. Connected to U10 with four wires and powered from the board.
Target module	Shelly X module X1, hardware V020, datasheet version 14 (attached). Same footprint as WBR3: 16 × 24 mm, 2 × 8 pads, 2 mm pitch.
Shelly product configuration	created in the Shelly X developer portal: data source “Tuya MCU” on IO4 (TX) / IO5 (RX), 9600 baud, 7 declared data points, 6 fields (state, current limit, mode, phase info, session energy, session duration), service script v3 written by AmperePoint.
Tools	multimeter (continuity and diode test on all 16 pads), module debug log read over Wi-Fi (frame level, every serial frame with timestamp), Python tools for MQTT and OCPP tests.



Shelly DevKit M1, top: USB-C, USB-UART bridge, RGB LED, Shelly-X-MOD1-H8 module with antenna



Bottom: pin labels. Pins 4 and 5 (TX, RX) go to the Q11; the pins labelled RX/TX are the USB/log port and are not used

Figure 1: The prototype module used in the tests.

3 Electrical connection

3.1 Module site U10: what the board actually uses

Measured with the charger switched off, diode test, red lead on the GND pad. Result: **only four pads have any connection to the controller**. All other pads show open circuit (OL) to GND, to VCC and to each other.

Pad	WBR3 name	On the Q11 board (measured)	Prototype (DevKit)	Target (Shelly X1 pad at the same position)
1	NC	no connection	not connected	RESET (module reset input, left open)
2	A_7	no connection	not connected	NTC (left open)
3	EN	no connection	not connected	NC
4	A_11	no connection	not connected	IO0
5	A_2	no connection	not connected	IO1
6	A_3	no connection	not connected	IO2
7	A_4	no connection	not connected	IO3

Pad	WBR3 name	On the Q11 board (measured)	Prototype (DevKit)	Target (Shelly X1 pad at the same position)
8	VCC	3.3 V from the board regulator; controller pins 1, 13, 19, 32, 48, 64	3V3 of the DevKit, through a removable jumper	3V3
9	GND	ground; controller pins 12, 18, 31, 47, 63	GND	GND
10	A_12	no connection	not connected	Sys btn (left open)
11	A_16	no connection	not connected	U0TXD (service UART, left open)
12	A_17	no connection	not connected	U0RXD (service UART, left open)
13	A_18	no connection	not connected	IO7
14	A_19	no connection	not connected	IO6
15	RXD	controller pin 16 = PA2, USART TX (controller transmits)	DevKit pin 5 = IO5, RX input, via 470 Ω	IO5 (RX)
16	TXD	controller pin 17 = PA3, USART RX (controller receives)	DevKit pin 4 = IO4, TX output, via 470 Ω	IO4 (TX)

Pad numbering follows the WBR3 datasheet: pad 1 at the dot printed next to “U10”, pads 1–8 up the right-hand column, pads 9–16 down the left-hand column (board viewed from the component side, antenna edge at the bottom). The X1 column is derived from the X1 datasheet (IO layout, page 5) and the WBR3 datasheet by overlaying the two footprints; it has not yet been verified on a physical X1 sample.

What this means for the drop-in replacement

- Power and ground of the X1 land on the same pads as on the WBR3.
- The serial link lands on X1 IO4 (TX) and IO5 (RX). This is exactly how the prototype is wired, and it works.
- The X1 pads RESET, NTC, Sys btn, U0RXD and U0TXD land on pads that the Q11 board leaves unconnected, so they float, which is acceptable for the X1.
- The WBR3 EN pad (module enable) is not driven by the board. The controller therefore cannot reset or disable the module; it did not need to during the tests.

3.2 Connections made for the prototype

Four wires between U10 and the DevKit (Fig. 2): VCC → jumper → 3V3, GND → GND, TXD → 470 Ω → DevKit pin 4 (IO4, TX), RXD → 470 Ω → DevKit pin 5 (IO5, RX). The 470 Ω resistors only protect against a wiring mistake (two outputs shorted: 3.3 V / 470 Ω $\approx$ 7 mA); with a 104 μ s bit time at 9600 baud they have no effect on the signal. The DevKit is powered exclusively from the board’s 3.3 V regulator; USB is disconnected while the jumper is in. Over the whole test period the module ran without a single brown-out reset, so the board regulator supplies the ESP32-C3 including Wi-Fi transmission.

3.3 Controller pins

Continuity from each used pad to the controller package (Fig. 3): VCC to pins 1, 13, 19, 32, 48 and 64; GND to pins 12, 18, 31, 47 and 63; pad RXD to pin 16; pad TXD to pin 17. The power and ground pattern is that of the STM32F103 / GD32F103 / GD32F303 family in LQFP64, and pins 16/17 are PA2/PA3, the USART2 pair of that family (USART1 in GD32 naming). The exact part number is hidden under the sticker; see question 1.

Wiring diagram: Shelly DevKit M1 → site U10 (WBR3 footprint) on the Q11 control board

Control board X2Q322_K_V4 (260703), controller X2Q3110W. Q11 board viewed from the component side, board edge with the module antenna at the bottom. DevKit viewed from the top, as in the photo. DevKit powered from the Q11 board through a removable jumper.

AMPERE POINT, 28 September 2026, EN v1 (from PL v2 of 24 September 2026).

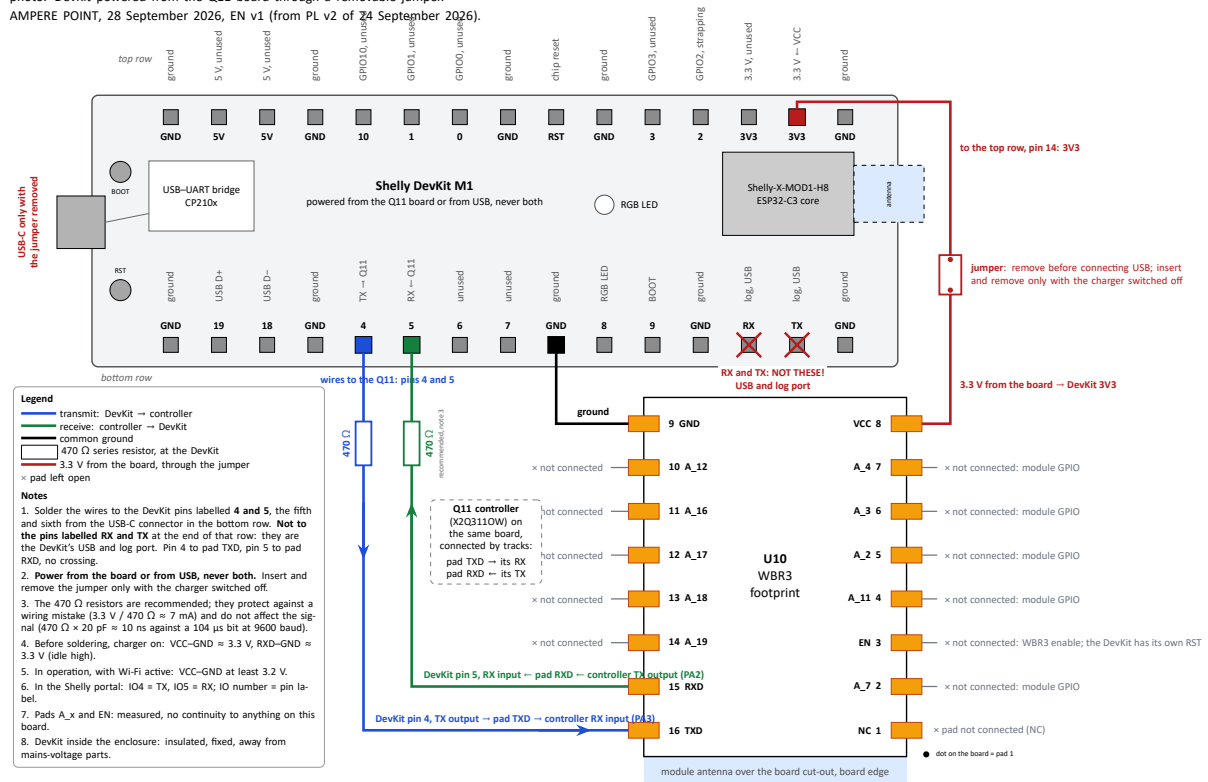


Figure 2: Wiring of the prototype (full-size PDF attached, page 2 shows every pad of U10).

Q11 controller: package pins and the U10 pads that reach them

Control board X2Q322_K_V4 (260703), controller with sticker "X2Q311OW OTA V22", LQFP64. View as in the photo: sticker readable, module site U10 on the right. Pin keys: u = top edge, d = bottom edge (numbered from the left), l = left edge, p = right edge (numbered from the bottom). Grey: standard pin number and function in the STM32F103 / GD32F103 / GD32F303 family. AMPERE POINT, 28 September 2026, EN v1 (from PL v2 of 25 September 2026): ground measured.

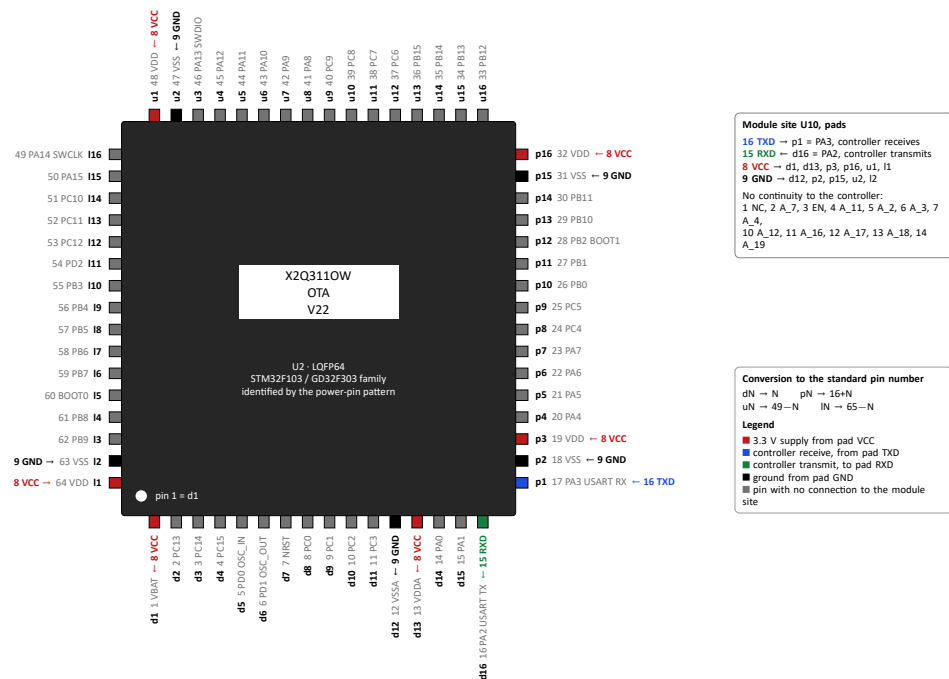


Figure 3: Controller pins with the U10 pads that reach them (full-size PDF attached).

4 Serial protocol as observed on the wire

4.1 Physical layer and frame format

9600 baud, 8 data bits, no parity, 1 stop bit, 3.3 V levels. Tuya MCU serial protocol: 55 AA | version | command | length (2 bytes, big-endian) | data | checksum (sum of all preceding bytes modulo 256). The controller sends version 03; the module sends version 00. The Shelly firmware implements this protocol natively (component "Tuya MCU"); AmperePoint wrote no protocol code.

4.2 Start-up handshake

Captured on 25 September 2026 at 11:51, when the module re-initialised after a configuration upload (the controller stayed powered):

Time	Direction	Frame (cmd, data)	Meaning
11:51:08	controller → module	0x24	controller asks for the Wi-Fi signal strength
11:51:08	module → controller	0x24 [C7]	answer: -57 dBm
11:51:09	module → controller	0x00	heartbeat
11:51:09	controller → module	0x00 [01]	heartbeat reply, 01 = controller already running (not a fresh boot)

Time	Direction	Frame (cmd, data)	Meaning
11:51:14	module → controller	0x03 [04]	Wi-Fi status 4 (connected to the cloud, in Tuya terms)
11:51:14	controller → module	0x03	acknowledged
11:51:14	module → controller	0x08	query all data points
11:51:16–17	controller → module	12 × 0x07	data point reports, in this order: 3, 4, 6, 7, 8, 9, 10, 14, 18, 19, 24, 13
11:51:16	controller → module	0x1C	controller asks for the time
11:51:17	module → controller	0x1C [01 1A 09 19 0B 33 12 05]	valid, 2026-09-25 11:51:18, weekday 5

The product-information exchange (commands 0x01 and 0x02) happens before the debug log is reachable over Wi-Fi and was not captured; see question 2.

4.3 Routine traffic

- Module → controller: heartbeat 0x00 every 15 s; Wi-Fi status 0x03 [04] every 30 s. The controller answers both.
- Controller → module: time request 0x1C and signal-strength request 0x24, each at irregular intervals of about 10–45 s. The module answers the time request only after it has synchronised its clock (NTP); before that it sends 01 00 00 00 00 00 00 00 (flag “valid” with a zero date). AmperePoint therefore provides a local NTP server when the charger is used without Internet.
- Controller → module, unsolicited 0x07 reports: temperature (DP 24) on every change, connection state (DP 13) on every change of the control-pilot state, current limit (DP 4) and switch (DP 18) after each write, session energy (DP 25) once at the end of a session.

4.4 Write behaviour

On a write (0x06) the controller first reports the **old** value, then the **new** value about one second later. Example, current limit 12 A → 10 A, 25 September 2026:

Time	Direction	Complete frame	Meaning
11:56:48	module → controller	55 AA 00 06 00 08 04 02 00 04 00 00 00 0A 21	set DP 4 (value type, 4 bytes) = 10
11:56:49	controller → module	55 AA 03 07 00 08 04 02 00 04 00 00 00 0C 27	report DP 4 = 12 (still the old value)
11:56:50	controller → module	55 AA 03 07 00 08 04 02 00 04 00 00 00 0A 25	report DP 4 = 10 (accepted)
10:52:37	module → controller	55 AA 00 06 00 05 12 01 00 01 01 1F	set DP 18 (bool) = 1, charging enabled
10:52:38	controller → module	55 AA 03 07 00 05 12 01 00 01 01 23	report DP 18 = 1
–	controller → module	55 AA 03 00 00 01 01 04	heartbeat reply, for reference

Values 8, 10, 12, 14 and 16 A were written and confirmed by the controller in the same way.

5 Data points

5.1 Three layers

The controller sends its data-point reports on the wire regardless of which module is listening; the Shelly module receives every frame exactly as the WBR3 did. What differs is where the meaning of a data point is defined: with the WBR3 it is the product definition in the Tuya cloud, with the Shelly module it is the product configuration written by AmperePoint. The log confirms that the Shelly module registers every reported data point, including those that are not declared in the product configuration (log entries such as `add int 24`, `add enum 13`, `add raw 19`).

Layer	Data points	Defined by
Received and stored by the module	all that the controller sends: 13 different IDs seen	the controller firmware
Declared in the Shelly product, available to the service script	7 (1, 3, 4, 6, 7, 8, 18)	AmperePoint, in the portal
Exposed as fields (app, local API, MQTT)	6 fields	AmperePoint, in the service script

Enumerations travel on the wire as an index byte; the names exist only in the product definition. The Shelly product uses the same order as the Tuya definition (question 7 asks for confirmation).

5.2 Comparison: Tuya product definition, controller behaviour, Shelly product

Definition taken from the Tuya product *Ampere Point Q11 PRO* (17 data points). “Reported” = seen in the module log on 25 September 2026: **dump** = in the answer to the query `0x08`, **spont.** = sent by the controller on its own, **never** = not seen in any frame.

DP	Tuya code	Type, unit, encoding	R/W	Reported by controller	In the Shelly product
1	forward_energy_total	value, kWh, scale 2 (123456 = 1234.56 kWh)	R	✗ never (not in dump, not spont.)	declared, meant for the lifetime counter; stays at 0 because nothing arrives. Question 4
3	work_state	enum, 8 states: 0 free, 1 insert, 2 free_fault, 3 wait, 4 charging, 5 pause, 6 end, 7 fault	R	✓ dump; values 0 and 3 seen	✓ field “mode”, index translated to name
4	charge_cur_set	value, A, 6–32	RW	✓ dump; writes 8/10/12/14/16 confirmed	✓ field “current limit”, limited to 6–16 A for the Q11

DP	Tuya code	Type, unit, encoding	R/W	Reported by controller	In the Shelly product
6, 7, 8	phase_a / b / c	raw, 7 bytes: voltage 2 B $\times$ 0.1 V, current 3 B $\times$ 0.001 A, power 2 B in W	R	✓ dump; 226–228 V seen while the contactor was closed; current and power not yet seen under load	✓ decoded in the script into the “phase info” field
9	power_total	value, kW, scale 3	R	✓ dump; 0	🔴 script sums the phases instead
10	fault	bitmap, 17 bits: ov_cr, ov2_cr_fault, ov_vol, undervoltage_alarm, contactor_adhesion, contactor_fault, earth_fault, meter_hardware_alarm, scram_fault, cp_fault, meter_commu_fault, card_reader_fault, cir_short_fault, adhesion_fault, self_test_alarm, leakagecurr_alarm, ov_Temp_fault	R	✓ dump; 0	🔴 received, not yet exposed as a field
13	connection_state	enum, 7: 0 12V, 1 12V_pwm, 2 9V, 3 9V_pwm, 4 6V, 5 6V_pwm, 6 error	R	✓ dump + spont.; values 0, 2, 4 seen	🔴 received, not yet exposed
14	work_mode	enum, 3: 0 charge_now, 1 charge_energy, 2 charge_schedule	RW	✓ dump; 0	🔴 received, not yet exposed
17	energy_charge	value, kWh, 1–200	RW	✗ never	not declared. Question 4
18	switch	bool	RW	✓ dump + spont. after write	✓ field “state”
19	local_timer	raw, 2 bytes: start hour, end hour	RW	✓ dump; 00 00	🔴 received, not yet exposed
23	system_version	string	R	✗ never	not declared. Question 4
24	temp_current	value, °C, –40–200	R	✓ dump + spont. on change; 18–20 °C	🔴 received, not yet exposed

DP	Tuya code	Type, unit, encoding	R/W	Reported by controller	In the Shelly product
25	charge_energy_once	value, kWh, scale 2	R	🟡 once: value 0, in one frame together with DP 18 = 0	🟡 session energy is currently computed in the script; will use DP 25
33	mode_set	raw	RW	✗ never	not declared. Question 5

The dump answering 0x08 contains 12 data points in this order: 3, 4, 6, 7, 8, 9, 10, 14, 18, 19, 24, 13. Data points 1, 17, 23 and 33 are absent from the dump and were never sent on their own.

6 Test results

Test	Result	Details
Communication with the controller	✓	Tuya MCU protocol, 9600 8N1; heartbeat answered; data points reported; no controller firmware change
Pads used by the board	✓	only VCC, GND, RXD, TXD; twelve pads open
Module powered from the board	✓	3.3 V from the board regulator; no brown-out during the tests
Charging on/off from the module	✓	DP 18 written, confirmed by the controller within 1 s
Current limit from the module	✓	8, 10, 12, 14, 16 A written and confirmed
Status read-out	✓	work state, connection state, temperature, fault bits, work mode, timer window, phase voltages
Time delivered to the controller	✓	0x1C answered with the correct local time once the module has NTP; a local NTP server is used off-line
Wi-Fi status to the controller	✓	0x03 [04] every 30 s; the charger display shows the network icon
Operation without any cloud	✓	the Shelly cloud disabled; module controlled over the local network (HTTP, WebSocket, MQTT) and over Bluetooth
Local integration layer	✓	see below
Charging under load	🟡	a vehicle simulator was used for the control-pilot states; a full session with current flowing is still to be done
Controller firmware update through the module	✗ not tested	see question 9
Wi-Fi reset from the charger menu	✗ not tested	see question 8

Local integration layer (middleware), for information

The Shelly module exposes a local API (HTTP, WebSocket, MQTT). On top of it AmperePoint runs a small software layer outside the charger: an MQTT broker, a local NTP server and a bridge that presents the Q11 to a charging back-end as an OCPP 1.6J charge point (registration, status, remote start, current limit via charging profile, meter values). All of this was tested against a local test server. It lives on a computer or server, not in the charger, and **needs nothing from the controller firmware**. It is mentioned here only so that the whole picture is visible.

7 Open questions for the factory

1. **Controller part number.** Which microcontroller is under the sticker “X2Q311OW OTA V22” (STM32F103xx, GD32F103xx, GD32F303xx, other), and what is its flash size?
2. **Product information frame.** What does the controller answer to the product query (0x01): product key *p*, firmware version *v*, module working mode *m*? Which Tuya MCU SDK / protocol version does the firmware implement? We could not capture this exchange because it happens before the module log is reachable.
3. **Unconnected pads.** Twelve pads of U10 (NC, A_7, EN, A_11, A_2, A_3, A_4, A_12, A_16, A_17, A_18, A_19) show no continuity to anything. Please confirm from the schematic that the board uses only VCC, GND, RXD and TXD of the module, and that the firmware does not expect any signal on the other pads (for example EN, or a reset line).
4. **Data points never reported.** DP 1 (forward_energy_total), DP 17 (energy_charge), DP 23 (system_version) and DP 33 (mode_set) were not in the dump and were never sent on their own. Under which conditions does the controller report each of them? DP 25 (charge_energy_once) was sent once, with value 0, together with DP 18 = 0: is that the end-of-session report?
5. **DP 33 mode_set.** What is the byte format and the meaning of this raw data point?
6. **DP 8.** In the export of Q21_11kW_with_OTA DP 8 is named charger_events, in the *Q11 PRO* definition it is phase_c. Our unit sends DP 8 as a 7-byte raw value in the same format as DP 6 and DP 7. Which definition does the shipped firmware follow, and does the format differ between firmware versions?
7. **Enumeration order.** Please confirm the index order of DP 3, DP 13 and DP 14 as listed in the table above.
8. **Wi-Fi status and pairing.** Which Wi-Fi status values (0x03, 0–7) does the controller use, and for what (display icon, behaviour)? What does the charger’s menu item for network reset / pairing send to the module (0x04 or 0x05)?
9. **Controller firmware update through the module.** Does the controller support the Tuya MCU update commands (0x0A start, 0x0B packet transfer)? Which file format is uploaded to the Tuya platform for the MCU OTA (plain binary?), and could the same file be provided to AmperePoint so that the Shelly module, which implements the same update commands, can update the controller? Does the bootloader tolerate an interrupted update?
10. **Time.** The controller asks for the time (0x1C) every 10–45 s. What does it use the time for (timer window DP 19, display, session records), and what happens if the module never supplies a valid time?
11. **Behaviour when the module is silent.** The controller kept working normally when the module sent nothing for a day (miswired prototype). Is there any timeout on the heartbeat or on the Wi-Fi status that changes the controller’s behaviour?

8 Attachments

- ShellyX_module_X1_datasheet_v14.pdf: datasheet of the target module (Shelly).
- schematic_wiring_DevKit_Q11_EN.pdf: wiring of the prototype and function of every pad of U10.
- schematic_Q11_MCU_pins_EN.pdf: controller pins with the U10 pads that reach them.
- Frame-level logs of the module (25 September 2026) are available on request.

AMPERE POINT — R&D · www.amperepoint.pl · Prepared on 28 September 2026 from measurements on a Q11 unit, the module debug log and the Tuya product definitions.